MTH102 – End-semester – 7th of May, 2026.

IMPORTANT: Give clear and **complete** explanations for everything!!!

1. (2 points) Show that if A and B are independent events then so are A^c and B^c.

2. (2 points) Consider a random variable for which $E[X] = 0$ and $\mathrm{Var}(X) = 1$. Then compute $E[(2+X)^2]$ and $\mathrm{Var}(4+3X)$.

3. (3 points) There are 10 coins in a box: one biased with a 0.7 probability that it will show a head when tossed, and the others unbiased. You pick a coin at random. To check if it is unbiased you toss it 1000 times. You get a head 650 times. What is the probability that you picked the unbiased coin?

4. (2 points) If we assume that the number of earthquakes in a year follows a Poisson distribution and that the probability of no earthquake in a year is 0.01, what is the probability that there will be 2 earthquakes in a year?

5. A biased coin, which gives a head with probability 0.0001, is tossed 10000 times. Assume that the tosses are independent events. Let X denote the random variable that represents the number of times you get a head. Find the probability that the number of heads are between 400 to 600 using the...

   (a) (2 points) binomial distribution

   (b) (2 points) Poisson approximation

   (c) (2 points) Central Limit Theorem

6. (2 points) Consider a normal distibution X with mean μ and standard deviation σ. Show that $P\{\mu - 2\sigma < X < \mu + 3\sigma\}$ is always the same no matter what μ and σ are.

7. (3 points) If X_1 and X_2 are independent and each a Poisson distribution, show that $X_1 + X_2$ is also a Poisson distribution.

8. (3 points) Show that if X is a binomial random variable with parameters n and p then $E[X^k] = npE[(Y+1)^{k-1}]$ where Y is also a binomial random variable but with different paramaters. What are those paramaters?

9. (2 points) Let $M(t)$ denote the moment generating function of random variable X. Show that $\left.\frac{d^2}{dt^2}\log(M(t))\right|_{t=0} = \mathrm{Var}(X)$.

10. Let $X_1, X_2, \ldots, X_n$ be *identical* and *independent* random variables each with mean 0, variance 1, and moment generating function, $M(t)$. Let $\bar{X}_n = \dfrac{X_1 + X_2 + \cdots + X_n}{n}$.

    (a) (2 points) What is the standard deviation, $\widetilde{\sigma}_n$, of $\bar{X}_n$?

    (b) (2 points) Derive an expression for the moment generating function, $\widetilde{M}_n(t)$, of $\bar{X}_n/\widetilde{\sigma}_n$ in terms of $M(t)$.

    (c) (3 points) Derive the moment generating function for the standard normal distribution.

    (d) (3 points) Show that $\lim_{n\to\infty} \widetilde{M}_n(t)$ is the moment generating function of the standard normal distribution.

11. (3 points) 1000 fair dice are rolled. Use the central limit theorem to estimate the probability of the sum being between 2000 and 3000.

12. (2 points) If X is a random variable with $\text{Var}(X) = 0$ and $E[X] = \mu$, what is $P\{\mu - 0.0001 < X < \mu + 0.0001\}$, i.e. what is the probability that the outcome is no more than 0.0001 away from the mean?